

$$3. \quad \vec{a} \cdot \vec{b} = |\vec{a}| \times |\vec{b}| \times \cos 30^\circ = 4 \times \sqrt{3} \times \frac{\sqrt{3}}{2} = 6$$

$$\begin{aligned} |\vec{a} + \vec{b}|^2 &= |\vec{a}|^2 + 2\vec{a} \cdot \vec{b} + |\vec{b}|^2 \\ &= 4^2 + 2 \times 6 + (\sqrt{3})^2 = 31 \end{aligned}$$

$$\text{故 } |\vec{a} + \vec{b}| = \sqrt{31}$$

$$4. \quad \vec{AB} = B - A = (5, -2) - (-1, 1) = (6, -3)$$

$$\vec{AC} = C - A = (1, 3) - (-1, 1) = (2, 2)$$

$$\vec{AB} \cdot \vec{AC} = (6, -3) \cdot (2, 2) = 6 \times 2 + (-3) \times 2 = 6$$

$$5. \quad \vec{a} \cdot \vec{b} = (1, k+1) \cdot (2k, 3)$$

$$= 1 \times 2k + (k+1) \times 3$$

$$= 2k + (3k+3) = 5k+3$$

$$\text{則 } 5k+3=18 \Rightarrow 5k=15 \Rightarrow k=3$$

$$6. \quad \because \vec{a} \perp \vec{b}$$

$$\therefore \vec{a} \cdot \vec{b} = 0$$

$$\text{而 } \vec{a} \cdot \vec{b} = (3, 4) \cdot (x, -9)$$

$$= 3 \times x + 4 \times (-9) = 3x - 36$$

$$\text{則 } 3x - 36 = 0 \Rightarrow 3x = 36 \Rightarrow x = 12$$

$$\because \vec{b} \parallel \vec{c}$$

$$\begin{aligned} \therefore \frac{x}{-8} &= \frac{-9}{y} \Rightarrow \frac{12}{-8} = \frac{-9}{y} \Rightarrow \frac{3}{-2} = \frac{-9}{y} \\ &\Rightarrow 3y = 18 \Rightarrow y = 6 \end{aligned}$$

$$\text{故 } y - x = 6 - 12 = -6$$

$$7. \quad \begin{vmatrix} 5 & 3 \\ 45 & 30 \end{vmatrix} = 5 \times 30 - 3 \times 45 = 150 - 135 = 15$$

$$\begin{vmatrix} 7 & -4 \\ 6 & 3 \end{vmatrix} = 7 \times 3 - (-4) \times 6 = 21 - (-24) = 45$$

$$\text{則 } m=15, n=45, \text{ 故 } m+n=15+45=60$$

$$8. \quad \vec{AB} = B - A = (98, 89) - (90, 80) = (8, 9)$$

$$\vec{AC} = C - A = (94, 87) - (90, 80) = (4, 7)$$

$$\triangle ABC \text{ 面積} = \frac{1}{2} \begin{vmatrix} 8 & 9 \\ 4 & 7 \end{vmatrix} = \frac{1}{2} \times |8 \times 7 - 9 \times 4|$$

$$= \frac{1}{2} \times |20| = \frac{1}{2} \times 20 = 10$$

$$9. \quad \text{由柯西不等式: } (2^2 + 3^2)(x^2 + y^2) \geq (2x + 3y)^2$$

$$\Rightarrow 13(x^2 + y^2) \geq 26^2 \Rightarrow x^2 + y^2 \geq 52$$

故 $x^2 + y^2$ 的最小值為 52

10. $\vec{a}$ 在 $\vec{b}$ 上的正射影:

$$\begin{aligned} \left(\frac{\vec{a} \cdot \vec{b}}{|\vec{b}|^2} \right) \vec{b} &= \frac{(12, 16) \cdot (7, 1)}{(\sqrt{7^2 + 1^2})^2} (7, 1) \\ &= \frac{12 \times 7 + 16 \times 1}{50} (7, 1) \\ &= 2(7, 1) = (14, 2) \end{aligned}$$

⑦ 單元 4 式的運算

(主題 1~10: 多項式的四則運算)

答案

1. 2

2. -15

3. $9x^4 - x^3$

4. $64x^3 - 140x^2 + 88x - 2$

5. 13

6. -7

7. 31

8. 7

9. $2x^3 + x^2 - 3x + 2$

10. 10

1. $\because f(x) = g(x)$

$\therefore$ 同次項的係數相等

$$\text{故 } -2 = b - 1, a + 1 = -1, 5 = c$$

$$\Rightarrow b = -1, a = -2, c = 5$$

$$\text{因此 } a + b + c = -2 + (-1) + 5 = 2$$

$$\begin{aligned} 2. \quad f(x) + g(x) &= (3x^2 - bx + c) + (ax^2 + 5x - 1) \\ &= (a+3)x^2 + (-b+5)x + (c-1) \end{aligned}$$

$\because f(x) + g(x)$ 為零多項式

$$\therefore a+3=0, -b+5=0, c-1=0$$

$$\text{故 } a=-3, b=5, c=1$$

$$\text{因此 } abc = -3 \times 5 \times 1 = -15$$

$$3. \quad f(x) = (3x^2)^2 - 8^2 = 9x^4 - 64$$

$$g(x) = (x-4)(x^2 + 4 \times x + 4^2) = x^3 - 4^3 = x^3 - 64$$

$$\text{故 } f(x) - g(x) = (9x^4 - 64) - (x^3 - 64) = 9x^4 - x^3$$

$$4. \quad f(x) = (2x)^2 - 2 \times 2x \times 5 + 5^2 = 4x^2 - 20x + 25$$

$$g(x) = (4x)^3 - 3 \times (4x)^2 \times 3 + 3 \times (4x) \times 3^2 - 3^3$$

$$= 64x^3 - 144x^2 + 108x - 27$$

$$f(x) + g(x)$$

$$= (4x^2 - 20x + 25) + (64x^3 - 144x^2 + 108x - 27)$$

$$= 64x^3 - 140x^2 + 88x - 2$$



$$\begin{array}{r}
 1 \quad +5 \quad +2 \\
 5. \quad 1+0-2 \overline{) 1 \quad +5 \quad +0 \quad +1 \quad -1} \\
 \underline{1 \quad +0 \quad -2} \\
 5 \quad +2 \quad +1 \\
 \underline{5 \quad +0 \quad -10} \\
 2 \quad +11 \quad -1 \\
 \underline{2 \quad +0 \quad -4} \\
 11 \quad +3
 \end{array}$$

餘式為 $11x+3$ ，故 $a=11$ ， $b=3$
 因此 $2a-3b=2 \times 11-3 \times 3=13$

6. 由除法原理

$$\begin{aligned}
 f(x) &= (x-3)(2x+1)+(-1) \\
 &= (2x^2-5x-3)-1=2x^2-5x-4
 \end{aligned}$$

故 $a=2$ ， $b=-5$ ， $c=-4$

因此 $a+b+c=2+(-5)+(-4)=-7$

7. 設 $f(x)$ 除以 x^2-5x+4 的商式為 $Q(x)$

$$f(x) = (x^2-5x+4)Q(x) + 6x+7$$

$$\begin{aligned}
 f(4) &= (4^2-5 \times 4+4) \times Q(4) + 6 \times 4+7 \\
 &= 0 \times Q(4) + 24+7=31
 \end{aligned}$$

8. 令 $x+2=0 \Rightarrow x=-2$

$$\begin{array}{r}
 1 \quad + \quad 1 \quad + \quad 0 \quad - \quad 2 \quad - \quad 5 \quad | \quad -2 \\
 \underline{- \quad 2 \quad + \quad 2 \quad - \quad 4 \quad + \quad 12} \\
 1 \quad - \quad 1 \quad + \quad 2 \quad - \quad 6 \quad | \quad +7
 \end{array}$$

故餘式為 7

9. $2x-1=2\left(x-\frac{1}{2}\right)$ ，令 $2x-1=0 \Rightarrow x=\frac{1}{2}$

$$\begin{array}{r}
 4 \quad + \quad 0 \quad - \quad 7 \quad + \quad 1 \quad + \quad 4 \quad | \quad \frac{1}{2} \\
 \underline{2 \quad + \quad 1 \quad - \quad 3 \quad - \quad 1} \\
 2 \quad | \quad 4 \quad + \quad 2 \quad - \quad 6 \quad - \quad 2 \quad + \quad 3 \\
 \underline{2 \quad + \quad 1 \quad - \quad 3 \quad - \quad 1}
 \end{array}$$

故商式 $Q(x)=2x^3+x^2-3x-1$ ，餘式 $R(x)=3$

因此 $Q(x)+R(x)=2x^3+x^2-3x+2$

10. 令 $x+1=0 \Rightarrow x=-1$

$$\begin{array}{r}
 2 \quad + \quad 1 \quad - \quad 5 \quad - \quad 3 \quad | \quad -1 \\
 \underline{- \quad 2 \quad + \quad 1 \quad + \quad 4} \\
 2 \quad - \quad 1 \quad - \quad 4 \quad | \quad 1 \rightarrow d \\
 \underline{- \quad 2 \quad + \quad 3} \\
 2 \quad - \quad 3 \quad - \quad 1 \rightarrow c \\
 \underline{- \quad 2} \\
 a \leftarrow 2 \quad - \quad 5 \rightarrow b
 \end{array}$$

因此 $abcd=2 \times (-5) \times (-1) \times 1=10$

⑧ 單元 4 式的運算

(主題 11~15：餘(因)式定理、方程式的解)

答案

- | | |
|-------|--------------------|
| 1. -3 | 2. 5 |
| 3. -1 | 4. 甲 |
| 5. 2 | 6. 21 |
| 7. 20 | 8. $\sqrt{21}$ |
| 9. -1 | 10. $\frac{19}{3}$ |

1. 令 $x+1=0$

$\Rightarrow x=-1$ ， $x=-1$ 代入多項式：

$$\begin{aligned}
 &3 \times (-1)^{201} + 4 \times (-1)^{100} - 5 \times (-1)^{52} + 1 \\
 &= 3 \times (-1) + 4 \times 1 - 5 \times 1 + 1 = -3 + 4 - 5 + 1 = -3
 \end{aligned}$$

故餘式為 -3

2. 令 $2x-1=0 \Rightarrow x=\frac{1}{2}$

$f(x)$ 除以 $2x-1$ 的餘式為

$$\begin{aligned}
 f\left(\frac{1}{2}\right) &= 80 \times \left(\frac{1}{2}\right)^4 + 56 \times \left(\frac{1}{2}\right)^3 - 36 \times \left(\frac{1}{2}\right)^2 + 2 \\
 &= 80 \times \frac{1}{16} + 56 \times \frac{1}{8} - 36 \times \frac{1}{4} + 2 \\
 &= 5 + 7 - 9 + 2 = 5
 \end{aligned}$$

3. 令 $x+1=0 \Rightarrow x=-1$

由餘式定理

$$\begin{aligned}
 \text{餘式：} &2 \times (-1)^3 - k \times (-1)^2 + 3 \times (-1) + 5 = 1 \\
 \Rightarrow &-2 - k - 3 + 5 = 1 \Rightarrow k = -1
 \end{aligned}$$

4. 甲：令 $x+2=0$

$$\Rightarrow x=-2, f(-2)=(-2+3)^{50}-1=1-1=0$$

乙：令 $x-2=0$

$$\Rightarrow x=2, f(2)=(2+3)^{50}-1=5^{50}-1 \neq 0$$

故甲是 $f(x)$ 的因式

5. 令 $x-3=0 \Rightarrow x=3$

由因式定理知， $3^3-6 \times 3^2+k \times 3+21=0$

$$\Rightarrow 27-54+3k+21=0$$

$$\Rightarrow 3k=6 \Rightarrow k=2$$

6.

$$\begin{array}{r}
 5 \quad + \quad 30 \quad - \quad 40 \quad - \quad 36 \quad + \quad 14 \quad | \quad -7 \\
 \underline{- \quad 35 \quad + \quad 35 \quad + \quad 35 \quad + \quad 7} \\
 5 \quad - \quad 5 \quad - \quad 5 \quad - \quad 1 \quad | \quad +21
 \end{array}$$

故 $f(-7)=21$

7. $x^2 - 8x - 84 = 0$

$$\left. \begin{array}{l} x \times 6 \\ x \times -14 \end{array} \right\} -14x + 6x = -8x$$

$$(x+6)(x-14) = 0 \Rightarrow x = -6 \text{ 或 } 14$$

取 $a = -6$, $b = 14$

故 $|a| + |b| = |-6| + |14| = 6 + 14 = 20$

8. 方程式的解 $x = \frac{-1 \pm \sqrt{1^2 - 4 \times 1 \times (-5)}}{2 \times 1} = \frac{-1 \pm \sqrt{21}}{2}$

$$\because a < b \therefore \text{取 } a = \frac{-1 - \sqrt{21}}{2}, b = \frac{-1 + \sqrt{21}}{2}$$

$$\begin{aligned} \text{故 } b - a &= \frac{-1 + \sqrt{21}}{2} - \frac{-1 - \sqrt{21}}{2} \\ &= \frac{-1 + \sqrt{21} - (-1 - \sqrt{21})}{2} \\ &= \frac{-1 + \sqrt{21} + 1 + \sqrt{21}}{2} = \frac{2\sqrt{21}}{2} = \sqrt{21} \end{aligned}$$

9. 兩根和: $\alpha + \beta = -\frac{-2}{1} = 2$

兩根積: $\alpha\beta = \frac{-1}{1} = -1$

$$\begin{aligned} (\alpha - 2)(\beta - 2) &= \alpha\beta - 2\alpha - 2\beta + 4 \\ &= \alpha\beta - 2(\alpha + \beta) + 4 \\ &= -1 - 2 \times 2 + 4 = -1 \end{aligned}$$

10. 兩根和: $\alpha + \beta = -\frac{-5}{1} = 5$, 兩根積: $\alpha\beta = \frac{3}{1} = 3$

$$\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta = 5^2 - 2 \times 3 = 19$$

$$\frac{\beta}{\alpha} + \frac{\alpha}{\beta} = \frac{\alpha^2 + \beta^2}{\alpha\beta} = \frac{19}{3}$$

⑨ 單元 4 式的運算

(主題 16~25: 複數、分式、根式)

答案

1. 22

2. -2

3. 6

4. $2 \pm \sqrt{3}i$

5. -6

6. -4

7. 7

8. 5

9. 11

10. -5

$$\begin{aligned} 1. (3-2i)(4+5i) &= 3 \times 4 + 3 \times 5i - 2i \times 4 - 2i \times 5i \\ &= 12 + 15i - 8i + 10 = 22 + 7i \end{aligned}$$

故實部為 22

$$2. i^{13} = i^{3 \times 4 + 1} = i^1 = i, i^{15} = i^{3 \times 4 + 3} = i^3 = -i$$

$$i^{16} = i^{4 \times 4} = 1, i^{18} = i^{4 \times 4 + 2} = i^2 = -1$$

則 $i^{13} + 2i^{15} + 3i^{16} + 4i^{18}$

$$= i + 2 \times (-i) + 3 \times 1 + 4 \times (-1)$$

$$= i - 2i + 3 - 4 = -1 - i$$

故 $a = -1$, $b = -1$, 因此 $a + b = -1 + (-1) = -2$

$$\begin{aligned} 3. a + bi &= \frac{15+5i}{2+i} = \frac{(15+5i) \times (2-i)}{(2+i) \times (2-i)} \\ &= \frac{30 - 15i + 10i + 5}{2^2 + 1^2} = \frac{35 - 5i}{5} \\ &= \frac{35}{5} - \frac{5}{5}i = 7 - i \end{aligned}$$

故 $a = 7$, $b = -1$, 因此 $a + b = 7 + (-1) = 6$

4. 方程式的解

$$\begin{aligned} x &= \frac{-(-4) \pm \sqrt{(-4)^2 - 4 \times 1 \times 7}}{2 \times 1} = \frac{4 \pm \sqrt{-12}}{2} \\ &= \frac{4 \pm \sqrt{12}i}{2} = \frac{4 \pm 2\sqrt{3}i}{2} \\ &= \frac{4}{2} \pm \frac{2\sqrt{3}}{2}i = 2 \pm \sqrt{3}i \end{aligned}$$

5. $\because 1+i$ 是方程式的根

$$\therefore 2(1+i)^2 + k(1+i) + 6 + 2i = 0$$

$$\Rightarrow 2 \times 2i + k(1+i) + 6 + 2i = 0$$

$$\Rightarrow k(1+i) = -6 - 6i$$

$$\Rightarrow k = \frac{-6 - 6i}{1+i} = \frac{-6(1+i)}{1+i} = -6$$

6. $\because$ 方程式有相等兩實根

$$\therefore \text{判別式} = 0$$

$$\Rightarrow (-4)^2 - 4 \times 1 \times (-k) = 0$$

$$\Rightarrow 16 + 4k = 0$$

$$\Rightarrow 4k = -16$$

$$\Rightarrow k = -4$$

$$\begin{aligned} 7. \frac{x}{x-2} - \frac{3x+2}{x^2-4} &= \frac{x(x+2)}{(x-2)(x+2)} - \frac{3x+2}{(x-2)(x+2)} \\ &= \frac{x^2 + 2x - 3x - 2}{(x-2)(x+2)} \\ &= \frac{x^2 - x - 2}{(x-2)(x+2)} = \frac{(x-2)(x+1)}{(x-2)(x+2)} \\ &= \frac{x+1}{x+2} \end{aligned}$$

故 $a = 2$, $b = 1$, 因此 $3a + b = 3 \times 2 + 1 = 7$

8. 方程式左、右兩側同乘 $(x+1)(x+3)$:

$$6(x+3) + x(x+1) = 2(x+1)(x+3)$$

$$\Rightarrow x^2 + x - 12 = 0 \Rightarrow (x+4)(x-3) = 0$$

$$\Rightarrow x = -4 \text{ 或 } 3$$

$$\therefore \alpha > \beta$$

$$\therefore \text{取 } \alpha = 3, \beta = -4$$

故 $3\alpha + \beta = 3 \times 3 + (-4) = 5$

9. 原式左、右兩側同乘 $(x-3)(x+2)$:

$$4x-7=A(x+2)+B(x-3)$$

(1) 令 $x=-2$ 代入 :

$$4 \times (-2) - 7 = A \times (-2+2) + B \times (-2-3)$$

$$\Rightarrow B=3$$

(2) 令 $x=3$ 代入 :

$$4 \times 3 - 7 = A \times (3+2) + B \times (3-3)$$

$$\Rightarrow A=1$$

$$\text{故 } 2A+3B=2 \times 1+3 \times 3=11$$

$$10. \because (a-b)(a^2+ab+b^2)=a^3-b^3$$

$$\therefore (\sqrt[3]{3}-2)(\sqrt[3]{9}+2\sqrt[3]{3}+4)$$

$$=(\sqrt[3]{3}-2)\left[(\sqrt[3]{3})^2+\sqrt[3]{3} \times 2+2^2\right]$$

$$=(\sqrt[3]{3})^3-2^3=3-8=-5$$

10 單元 5 直線與圓

(主題 1~13: 直線方程式)

答案

1. $m_1 < m_3 < m_2$

2. 1

3. $2x-y-10=0$

4. $2x+3y+3=0$

5. 9

6. -49

7. $-\frac{7}{3}$

8. $2x-y+1=0$

9. 2

10. 36

1. $m_1 = \frac{3-(-3)}{\frac{1}{2}-\frac{7}{2}} = \frac{6}{-3} = -2$, $m_2 = -\frac{1}{-3} = \frac{1}{3}$

$\therefore y=3$ 是水平線 $\therefore$ 其斜率 $m_3=0$

$\therefore -2 < 0 < \frac{1}{3} \therefore m_1 < m_3 < m_2$

2. $A、B$ 連線斜率 $= \frac{2-6}{10-(-2)} = \frac{-4}{12} = -\frac{1}{3}$

$B、C$ 連線斜率 $= \frac{(a+4)-2}{a-10} = \frac{a+2}{a-10}$

$\therefore A、B、C$ 在同一直線上

$\therefore \frac{-1}{3} = \frac{a+2}{a-10} \Rightarrow 3a+6 = -a+10$

$\Rightarrow 4a=4 \Rightarrow a=1$

3. 由點斜式, $y-(-2)=2 \times (x-4)$

$\Rightarrow y+2=2x-8 \Rightarrow -2x+y+10=0$

$\times(-1)$

$\Rightarrow 2x-y-10=0$

4. 由斜截式, $y = -\frac{2}{3}x + (-1)$

$\Rightarrow 3y = -2x - 3 \Rightarrow 2x + 3y + 3 = 0$

5. $\therefore$ 直線 L 的 x 截距為 3

$\therefore$ 直線 L 通過點 $(3,0)$

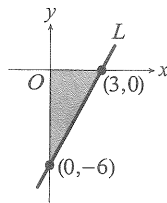
由點斜式

$L: y-0=2 \times (x-3)$

$\Rightarrow 2x-y-6=0$

x	3	0
y	0	-6

因此所求三角形之面積 $= \frac{1}{2} \times 3 \times 6 = 9$



6. (1)

x	28	0
y	0	7

 x 截距 $a=28$, y 截距 $b=7$

(2) 直線 $L: x+4y=28$ 的斜率 $m = -\frac{1}{4}$

故 $abm = 28 \times 7 \times \left(-\frac{1}{4}\right) = -49$

7. (1) 直線 L 通過 $A、B$ 兩點

則斜率 $= \frac{2k-2}{1-(-k)} = \frac{2k-2}{k+1}$

(2) 直線 $L_2: x+5y-5=0$ 的斜率 $= -\frac{1}{5}$

$\therefore L \perp L_2 \therefore \frac{2k-2}{k+1} \times \left(-\frac{1}{5}\right) = -1$

$\Rightarrow 5k+5=2k-2 \Rightarrow 3k=-7$

$\Rightarrow k = -\frac{7}{3}$

8. 設直線 $L: 4x-2y+k=0$

點 $(0,1)$ 代入 $L: 4 \times 0 - 2 \times 1 + k = 0 \Rightarrow k=2$

故直線 $L: 4x-2y+2=0 \xrightarrow{\div 2} 2x-y+1=0$

9. (1) 直線 $\overleftrightarrow{BC}$ 的斜率 $= \frac{3-(-2)}{-7-5} = -\frac{5}{12}$

(2) $\overleftrightarrow{BC}: y-(-2) = -\frac{5}{12}(x-5)$

$\Rightarrow 5x+12y-1=0$

(3) 點 $A(3,1)$ 到 $\overleftrightarrow{BC}$ 的距離

$= \frac{|5 \times 3 + 12 \times 1 - 1|}{\sqrt{5^2 + 12^2}} = \frac{26}{13} = 2$

10. 設 $L_1: 3x+4y+k_1=0$, $L_2: 3x+4y+k_2=0$

點 $(-29,23)$ 代入 $L_1: 3 \times (-29) + 4 \times 23 + k_1 = 0$

$\Rightarrow k_1 = -5$

點 $(31,23)$ 代入 $L_2: 3 \times 31 + 4 \times 23 + k_2 = 0$

$\Rightarrow k_2 = -185$